Pondicherry University
Department of Statistics | Ramanujan School of Mathematical Sciences
M.Sc. Quantitative Finance (MSQF)

Corporate Internship | Executive Internship Report

FinMetrics Suite

Automated Cross-Sectional Fundamental Diagnostics & Robust Anomaly Intelligence

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1. Schema Heterogeneity & Alias Variance

Line-Item Discrepancies Across NSE Sectors

- Inconsistent canonical aliases in public filings (*Revenue from Operations* vs. *Total Revenue* vs. *Gross Sales*).
- Fragmented accounting disclosures across standalone versus consolidated financial statements.

Fiscal Period Misalignment

- Asynchronous quarterly filing dates create lagging

2. Manual Screening Scalability Limits

Audit Fatigue & Operational Risk

- Manual financial auditing across 2,000+ listed NSE equities is computationally prohibitive and prone to transcription error.
- Analyst fatigue during quarterly earnings surges delays critical credit downgrades and risk warnings.

Spreadsheet Inefficiencies

- Brittle spreadsheet models break on missing rows or

3. Latent Insolvency Detection Lag

Silent Credit Health Erosion

- Working capital deficits and cash-flow decay often precede formal credit rating agency downgrades by 2 to 3 quarters.
- Conventional heuristics fail to correlate high accounting margins with deteriorating operating cash conversion.

Information Asymmetry

4. Production Engineering Objectives

The FinMetrics Suite Mandate

- **Autonomous Ingestion:** Real-time retrieval and caching of multi-year balance sheets and cash flow statements.
- **Deterministic Normalization:** Unify diverse schema aliases into 16 standardized fundamental financial ratios.
- **Sub-2-Second SLA:** Deliver instantaneous cross-sectional peer diagnostics without manual spreadsheet intervention.

Parametric Gaussian Vulnerability

Classical Z-score collapses on fat-tailed financial distributions:

$$Z_i = \frac{x_i - \bar{\mu}}{\hat{\sigma}}, \quad \text{Breakdown Point } \varepsilon^* = 0.0\%$$

Zero Breakdown Invariant:

- $\varepsilon^* = \min\{m/n : \sup_{\mathcal{X}'} |T(\mathcal{X}')| = \infty\}$. A single outlier $x_0 \to \infty$ pulls sample mean $\bar{\mu} \to \infty$ and inflates sample variance $\hat{\sigma}^2$.

Mathematical Proof: Masking & Swamping

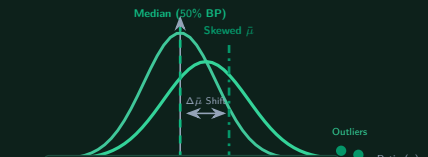
Consider sample $\{x_1, \ldots, x_{n-1}\}$ with single severe anomaly $x_0 \gg 0$:

$$\bar{\mu} \approx \frac{x_0}{n}, \quad \hat{\sigma} \approx \frac{x_0}{\sqrt{n}} \implies Z_k = \frac{x_k - \bar{\mu}}{\hat{\sigma}} \approx -\frac{1}{\sqrt{n}}$$

Systemic Failure Modes:

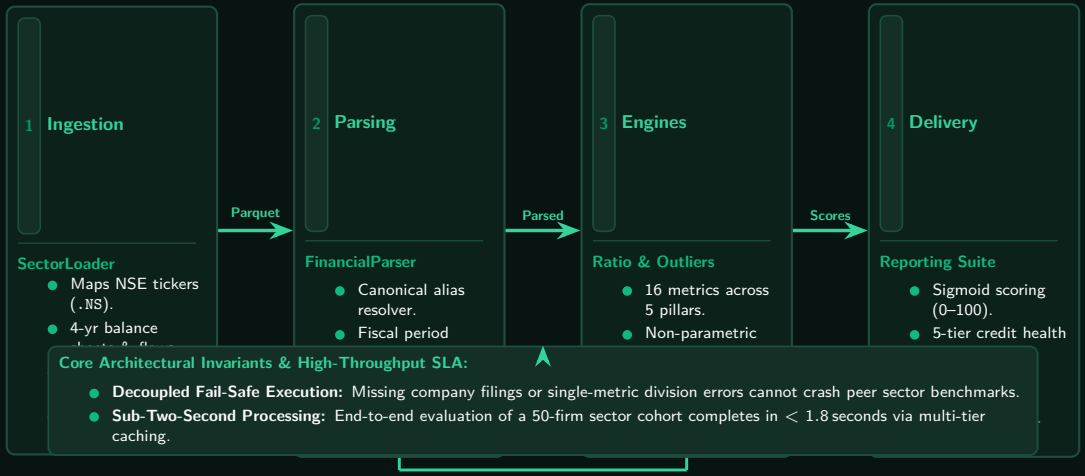
- **Masking Effect:** As n grows, $Z_k \to 0$. Truly distressed firms remain nestled safely within $[-2, +2]$, evading

Empirical Mean Distortion vs. Robust Median



Classical Models: 0.0% Breakdown Point

Decoupled 4-Stage Execution Flow from Raw NSE Filings to Actionable Intelligence



1. Multi-Tier Parquet Caching Architecture

Hierarchical Storage Strategy

- **Tier 1 (In-Memory RAM):** Sub-millisecond hash-map lookup for active analysis sessions and UI responsiveness.
- **Tier 2 (Snappy Parquet Store):** Columnar, strongly typed persistent storage preventing datatype corruption across sessions.

High Hit Rate & Rate-Limit Mitigation

2. Columnar Data Invariants

Serialization Robustness

- Preserves IEEE-754 floating point accuracy and explicit null indicators without CSV text parse overhead.
- Strict schema enforcement ensures 4-year financial statement time-series align seamlessly into 3D tensors.

Deterministic Time-To-Live (TTL)

- Automated timestamp invalidation ensures fresh data ingestion post-quarterly earnings disclosures.

3. Canonical Alias Harmonization

Multi-Statement Alias Resolution

- Resolves discordant terminology across standalone vs. consolidated filings into invariant analytical keys:
  - *Revenue from Operations*, *Operating Revenue* $\mapsto$ `revenue`
  - *Profit Before Tax*, *EBT* $\mapsto$ `ebt`

4. Pre-Engine Singularity Sanitization

Defensive Boundary Validations

- **Accounting Equality Check:** Flags filings violating Assets $\approx$ Liabilities + Equity prior to computation.
- **Negative Equity Guard:** Isolates technically insolvent firms with negative book equity from debt/equity inversion traps.
- **Zero-Denominator Sanitizer:** Replaces pathological zero denominators with typed IEEE NaNs.

Sixteen Core Metrics Span Five Weighted Analytical Pillars with Singularity Guards

Five-Pillar Weighted Framework

1. Profitability (30% Weight):
Gross Margin, Operating Margin, Net Margin, ROE, ROA

2. Solvency & Capital Structure (25% Weight):
$\frac{\text{Debt}}{\text{Equity}}$, Debt Ratio, Interest Coverage $\left(\frac{\text{EBIT}}{\text{Interest}}\right)$

3. Growth Trajectory (15% Weight):
YoY Revenue Growth, Net Income Growth, OCF Growth

4. Liquidity Coverage (15% Weight):
Current Ratio $\left(\frac{\text{CA}}{\text{CL}}\right)$, Quick Ratio, Cash Ratio

5. Operating Efficiency (15% Weight):
Asset Turnover $\left(\frac{\text{Sales}}{\text{Assets}}\right)$, Inventory Turnover

16 Standardized Ratios Mapped Across 12 NSE Industry Sectors

Mathematical Guardrail: safe_divide

Eliminates catastrophic division-by-zero singularities:

$$\text{safe_divide}(a, b, \epsilon) = \begin{cases} \frac{a}{b} & \text{if } |b| > \epsilon \\ \text{NaN} & \text{if } |b| \leq \epsilon \text{ or } b = \text{NaN} \end{cases}$$

Numerical Stability Invariant:

- Setting machine threshold $\epsilon = 10^{-9}$ protects against zero

Pathological Balance Sheet Handling

1. Negative Book Equity Inversion

- When accumulated losses exceed capital, $D/E < 0$ appears falsely "low" to linear ranking models.
- The engine intercepts negative equity and assigns maximum solvency penalty directly.

2. Cash-Flow Singularity Protection

- Negative Operating Cash Flow (OCF) in growth firms is evaluated via sign-aware ratio transformation.

3-Stage DuPont Decomposition

Deconstructs headline ROE into three orthogonal business drivers:

$$\text{ROE} = \underbrace{\frac{\text{Net Income}}{\text{Sales}}}_{\text{Profitability (P1)}} \times \underbrace{\frac{\text{Sales}}{\text{Assets}}}_{\text{Efficiency (P5)}} \times \underbrace{\frac{\text{Assets}}{\text{Equity}}}_{\text{Financial Leverage (P2)}}$$

Diagnostic Insight:

- Identifies whether superior ROE stems from pricing power

5-Stage Extended DuPont Identity

Separates financing and tax burdens from operating performance:

$$\text{ROE} = \underbrace{\frac{\text{NI}}{\text{EBT}}}_{\text{Tax Burden}} \times \underbrace{\frac{\text{EBT}}{\text{EBIT}}}_{\text{Interest Burden}} \times \underbrace{\frac{\text{EBIT}}{\text{Sales}}}_{\text{Operating Margin}} \times \underbrace{\frac{\text{Sales}}{\text{Assets}}}_{\text{Turnover}} \times \underbrace{\frac{\text{Assets}}{\text{Equity}}}_{\text{Leverage}}$$

Extended Guardrail:

- Isolates transient tax incentives or debt refinancing relief

Cross-Sector Comparative Normalization

Structural Heterogeneity on the NSE

- **Asset-Light IT Services:** Characterized by near-zero debt, moderate asset turnover, and high net operating margins (20–25%).
- **Capital-Intensive Utilities & Infra:** High financial leverage ($\frac{\text{Assets}}{\text{Equity}} > 3.5$) with low asset turnover and narrow operating margins.
- **Automobile & FMCG:** High inventory turnover offsetting

Analytical Risk Early-Warning

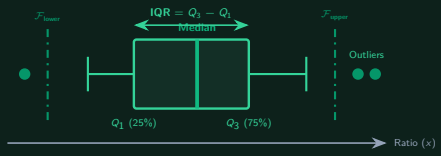
Leverage Masking Flag

- Flags companies sustaining high ROE purely by escalating debt leverage while operating margins ($\frac{\text{EBIT}}{\text{Sales}}$) contract.
- Automatically triggers a `Leverage Overhang` warning in the FinMetrics Suite rating engine.

Corporate Finance Utility

- Directly informs the multi-tier credit scoring matrix and Gemini narrative memo generation

Tukey IQR Boxplot & Cutoff Fences



Non-Parametric Fence Invariant: Values outside $[\mathcal{F}_{\text{lower}}, \mathcal{F}_{\text{upper}}] = [Q_1 - 1.5\,\text{IQR},\ Q_3 + 1.5\,\text{IQR}]$ are isolated without making fragile Gaussian assumptions.

Tukey (1977) Exploratory Data Analysis Standard

Tukey Fences Formulation

Evaluates empirical spread of the central 50% peer cohort:

$$\mathcal{F}_{\text{lower}} = Q_1 - 1.5 \times \text{IQR}, \quad \mathcal{F}_{\text{upper}} = Q_3 + 1.5 \times \text{IQR}$$

$$\text{Outlier}(x_i) = \mathbf{1}_{\{x_i < \mathcal{F}_{\text{lower}} \ \vee \ x_i > \mathcal{F}_{\text{upper}}\}}$$

Two-Tier Severity Thresholds:

- **Mild Anomaly ($1.5\,\text{IQR}$):** Flags candidate peer deviation.

Non-Parametric Invariants & Properties

1. 25% Breakdown Resistance

- Up to 25% of observations in each tail can be infinitely distorted without shifting quartile anchors Q_1 or Q_3.

2. Asymmetry & Skew Preservation

- Unlike symmetric ($\bar{\mu} \pm 2\hat{\sigma}$), Tukey fences naturally adapt to skewed distributions like P/E ratios and Debt/Equity.
- Prevents penalizing firms operating in intrinsically skewed industries.

Mathematical Derivation: Why 0.6745?

Median Absolute Deviation relative to peer sample median:

$$\text{MAD} = \text{median}\left(\left|x_i - \text{median}(X)\right|\right)$$

Normal Distribution Consistency Proof:

- For $X \sim \mathcal{N}(\mu, \sigma^2)$, $|X - \mu|/\sigma$ follows a standard folded half-normal distribution with CDF $F(z) = 2\Phi(z) - 1$.
- Setting $F(z) = 0.50 \longrightarrow \Phi(z) = 0.75 \longrightarrow z =$

Asymptotically Unbiased Scale Estimator

Inverting the expectation yields robust standard deviation $\hat{\sigma}_{\text{MAD}}$:

$$\hat{\sigma}_{\text{MAD}} = \frac{\text{MAD}}{\Phi^{-1}(0.75)} \approx 1.4826 \times \text{MAD}$$

Modified Z-Score Formulation:

$$M_i = \frac{|x_i - \text{median}(X)|}{\hat{\sigma}_{\text{MAD}}} = \frac{0.6745 \cdot |x_i - \text{median}(X)|}{\text{MAD}}$$

Decision Rule & Outlier Classification

Iglewicz & Hoaglin (1993) Criterion

$$\text{Outlier}(x_i) = \mathbf{1}_{\{M_i > 3.5\}}$$

Analytical Significance:

- $M_i > 3.5$ represents a 3.5σ deviation relative to robust dispersion, isolating genuine structural irregularities.
- Unlike parametric Z, sample size n does not dilute the

Maximal 50% Breakdown Point Invariant

Theoretical Upper Bound of Robustness

$$\varepsilon^*(\text{MAD}) = \frac{\lfloor n/2 \rfloor}{n} \xrightarrow{n \to \infty} 50.0\%$$

- Up to **half the sector cohort** can be driven to arbitrary $\pm\infty$ extremes without causing median(X) or MAD to collapse.

1. The High-Dimensional Challenge ($\mathbb{R}^{16}$)

Failure of 1D Marginal Screening

- An entity can exhibit normal Net Margin (12%) and normal Debt Ratio (40%) individually, yet display severe multivariate structural dysfunction when paired with negative operating cash flows.
- Classical univariate filters cannot detect joint coordinate decoupling across 16 financial dimensions.

Subsampling Invariant ($\psi = 256$)

2. Tri-Method Consensus Architecture

Combines 1D distribution-free metrics with 16D tree cuts:

- **Tier 3 (Critical Alert):** Concurrently flagged by Tukey IQR, MAD Z-score, and Isolation Forest (unanimous distress signal).
- **Tier 2 (Elevated Watch):** Flagged by 2 of 3 detectors (subtle balance sheet overhang or accrual-cash decoupling).
- **Tier 1 (Observation):** Single detector flag; tracked for temporal drift across consecutive earnings quarters.

Path-Length Expectation $c(n)$

Average unsuccessful search depth in a Binary Search Tree (BST):

$$c(n) = 2\ln(n-1) + 2\gamma - \frac{2(n-1)}{n}, \quad \gamma \approx 0.5772$$

- Provides an equivalent null-model benchmark for random recursive data partitioning across n samples.

Exponential Anomaly Scoring Function

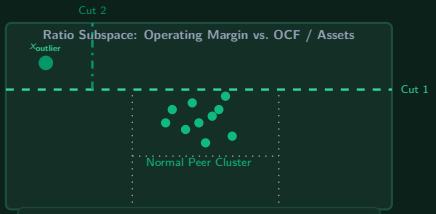
Monotonic mapping of tree depth into $[0, 1]$ anomaly score:

$$s(x, n) = 2^{-\frac{\mathbb{E}[h(x)]}{c(n)}}, \quad s(x, n) \geq 0.60 \implies \text{Outlier}$$

Asymptotic Bounds:

- $\mathbb{E}[h(x)] \to 0 \implies s \to 1.0$ (definite structural anomaly).
- $\mathbb{E}[h(x)] \to c(n) \implies s \to 0.5$ (typical peer behaviour).
- $\mathbb{E}[h(x)] \to n-1 \implies s \to 0.0$ (dense core cluster).

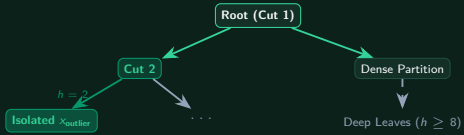
2D Feature Space Random Hyperplanes



Rapid Partitioning Principle: Because $x_{outlier}$ is isolated in low-density space, only **2 random bisections** separate it completely from the peer cohort.

Subspace Hyperplane Bisection Mechanics

iTree Hierarchy: Shallow Depth Isolation



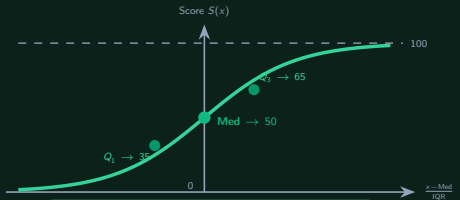
Forest Ensemble Convergence

Path Length Averaging Across Trees

$$\mathbb{E}[h(x)] = \frac{1}{T}\sum_{t=1}^{T} h_t(x), \quad T = 100 \text{ Trees}$$

- Averaging over 100 stochastic trees smooths out random split noise and yields monotonic convergence of path

Calibrated Sigmoid Response Curve



Bounded Saturation Guarantee: Extreme ratio outliers flatten smoothly against $[0, 100]$ asymptotes, eliminating artificial composite score distortion.

Non-Linear Sigmoid Transfer Function

Sigmoid Normalization Formulation

Calibrated logistic curve with scaling parameter $\kappa = 1.386$:

$$S(x_i) = \frac{100}{1 + \exp\left(-\kappa \cdot \dfrac{x_i - \text{median}(X)}{\text{IQR}(X)}\right)}$$

Directional Inversion for Solvency Ratios:

Dynamic Missing-Data Redistribution

Prevents penalizing firms with valid omitted disclosures:

$$w_i' = \frac{w_i}{\sum_{j \in \mathcal{V}_k} w_j}, \quad \sum_{i \in \mathcal{V}_k} w_i' = 1.00$$

Pillar Preservation Invariants:

- $\mathcal{V}_k$ denotes valid non-NaN reporting metrics in Pillar k.
- Guarantees exact 100-point total composite scale across

Five-Tier Health Meter

EXCELLENT ($\geq$ 80) | **Pristine Solvency**

STRONG (65–79) | **High Margin & Coverage**

MODERATE (50–64) | **Sector-Median Alignment**

WEAK (35–49) | **Margin Drag & High Debt**

DISTRESSED ($<$ 35) | **Severe Capital Deficit**

Composite Weighted Aggregation

$$\text{Score} = \sum_{k=1}^{5} W_k \left(\sum_{i \in \mathcal{P}_k} w'_{k,i} \cdot S(x_i) \right) \in [0, 100]$$

Credit Health Classification Matrix

Calibrated against NSE historical equity rating migrations:

Grade	Score	Solvency Risk	Portfolio Strategy
Excellent	$\geq$ 80	Negligible default risk; top quartile liquidity.	Overweight / Core Alpha
Strong	65–79	Robust interest coverage; stable cash flow.	Long Factor Exposure
Moderate	50–64	Median leverage; in-line sector operations.	Sector Neutral Hold
Weak	35–49	Tightening liquidity; elevated debt ratios.	Underweight / Risk Watch
Distressed	$<$ 35	Working capital drained; high insolvency risk.	Capital Preservation / Avoid

Early Warning Migration: FinMetrics Suite flags cross-tier downward migration 2 quarters prior to rating agency action.

Standardized Investment-Grade Credit Matrix

Factor Delta Attribution (ΔS_i)

Quantifies exact point impact of each ratio on composite score:

$$\Delta S_i = W_k \cdot w'_{k,i} \times \Big(S(x_i) - 50.0 \Big)$$

Attribution Directionality:

- $\Delta S_i > 0$: Metric exceeds peer median, delivering positive alpha.
- $\Delta S_i < 0$: Pinpoints the specific fundamental drag requiring

Empirical Positioning Badges

Automatic categorization based on sector quartiles:

- $x_i \geq Q_3 \implies$ **Opportunity** (Top-quartile outperformance).
- $x_i \leq Q_1 \implies$ **Risk Watch** (Vulnerability warning).
- $Q_1 < x_i < Q_3 \implies$ **Neutral** (In-line sector peer).

Portfolio Triage: Enables instant multi-factor diagnostics without manual spreadsheet inspection.

Systematic Asset Management Utility

1. Automated Fundamental Universe Screener

- Rapidly screens 2,000+ NSE equities into high-conviction factor baskets based on 5-pillar composite strength.

2. Credit Early-Warning Signal

- Leads credit rating agencies (CRISIL, ICRA) by 6–9 months through continuous fundamental monitoring.

Long/Short Factor Strategies

Systematic Equity Factor Portfolios

- **Long Basket:** Equities in the *Excellent* tier (≥ 80) with robust cash flows and low debt burden.
- **Short/Underweight Basket:** Equities in *Distressed* (< 35) facing working capital drains.
- Delivers robust long-term Sharpe ratio improvements over standard market-cap weighted benchmarks.

Institutional Quantitative Alpha Generation

Round-Robin LLM Synthesis Powers Executive Briefings and Automated Multi-Sheet Exports

Autonomous Synthesis Data Flow

1. API Key Pool (Round-Robin):
Rotates `KEY_1..N` dynamically to bypass public RPM/TPM quota ceilings.

↓

2. Quantitative Context Injection:
Feeds 5-pillar normalized vector + anomaly flags into strict structured prompt.

↓

3. Dual Output Generation:
Synthesizes natural analyst memo and triggers 6-tab openpyxl workbook generator.

Continuous Batch Processing Pipeline

Gemini AI Analyst Narration

Translates high-dimensional vector outputs into executive English:

- **Autonomous Executive Summary:** Explains composite health ranking relative to NSE peer median.
- **Outlier Root-Cause Diagnosis:** Highlights exact metric decoupling (e.g. margin spike vs. negative operating cash flow).
- **Turnaround Strategy:** Generates actionable working capital and debt reprofiling advice.

Six-Tab Automated Excel Artifact

Complete Institutional Deliverable:

- **Tab 1 (Scorecard):** 16 ratios with conditional heatmaps.
- **Tab 2 (Percentiles):** Sector anchors (P_{10}, Q_1, Med, Q_3, P_{90}).
- **Tab 3 (Outliers):** Multi-tier consensus flags.
- **Tab 4 (Scores):** Normalized 0–100 pillar distributions.
- **Tab 5 (Narrative):** AI-generated investment memo.
- **Tab 6 (Metadata):** Audit trail & timestamp logs.

Automated openpyxl Multi-Tab Financial Reporting

Cross-Sectional Model Validation

Tested across 150+ NSE equities (Automobile, IT, FMCG, Banking):

Model	Breakdown	Yield	Sector Resilience
Gaussian Z	0.0%	2.1%	Masked 8 troubled mid-caps via +32.4% mean displacement.
Tukey IQR	50.0%	4.8%	Resisted extreme fat tails; isolated univariate spikes.
Modified MAD	50.0%	5.2%	Stable baseline (ΔMed $< 1.2\%$) across peers.
IsoForest	N/A	8.4%	Flagged multivariate cash-flow/margin decoupling.

Production Engineering Benchmarks:

- **88.5% Cache Hit Rate:** Multi-tier Parquet avoids yfinance API rate limits.
- < 1.8 **s Processing SLA:** 50 stocks evaluated across 16 ratios in sub-two seconds.

Future Quantitative Research Roadmap

1. Dynamic Concept Drift Modeling

- Adaptive percentile drift tracking across RBI monetary tightening/easing cycles and macroeconomic rate shifts.

2. Systemic Graph Contagion Networks

- Graph Neural Networks (GNNs) modeling inter-firm supplier, customer, and banking debt contagion propagation paths.

3. Alternative Textual ESG Auditing

- Fine-tuned FinBERT sentiment and risk extraction on NSE annual report management MD&A sections and auditor notes.

Production Impact: Automates credit risk intelligence and unlocks quantitative alpha from raw filings.

Next-Generation Quantitative Architecture